C13 Halogenoalkanes – Knowledge Organiser: Self-Test

AQA A-level Chemistry • Set 1: Nucleophilic Substitution **Paper 2**

#	Question	Attempt 1	Attempt 2	Attempt 3	Answer
1	Polarity of the C–X bond				$C^{\delta+}–X^{\delta-}$
2	Nucleophile				Electron pair donor
3	Three nucleophiles in the AQA spec				OH^-, CN^-, NH_3
4	Trend in C–X bond enthalpy down Group 7				Decreases
5	Most reactive halogenoalkane				Iodoalkane
6	Why iodoalkanes are most reactive				Weakest C–X bond
7	Reagent and conditions to make an alcohol				NaOH(aq), warm
8	Reagent and conditions to make a nitrile				KCN in ethanol/water, reflux
9	Reagent and conditions to make an amine				Excess conc NH_3 in ethanol, heat in sealed tube
10	Product of bromoethane with KCN				Propanenitrile

Fold the Answer column behind the page. Answer each question from memory, then unfold and check. Repeat on a later day in the next Attempt column.

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AQA A-level Chemistry • Set 2: Elimination and Classification **Paper 2**

#	Question	Attempt 1	Attempt 2	Attempt 3	Answer
1	Reagent and conditions for elimination				KOH in ethanol, hot
2	Organic product of elimination				Alkene
3	Role of OH^- in elimination				Base
4	Role of OH^- in substitution				Nucleophile
5	Reaction of halogenoalkane with $OH^-(aq)$				Nucleophilic substitution
6	Reaction of halogenoalkane with OH^- in ethanol				Elimination
7	Product of 2-bromopropane with ethanolic KOH				Propene
8	Primary halogenoalkane				C–X carbon bonded to one other carbon
9	Secondary halogenoalkane				C–X carbon bonded to two other carbons
10	Tertiary halogenoalkane				C–X carbon bonded to three other carbons

Fold the Answer column behind the page. Answer each question from memory, then unfold and check. Repeat on a later day in the next Attempt column.

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AQA A-level Chemistry • Set 3: Ozone Depletion and Products **Paper 2**

#	Question	Attempt 1	Attempt 2	Attempt 3	Answer
1	Formula of ozone				O_3
2	Role of the ozone layer				Absorbs UV radiation
3	Compounds that deplete ozone				CFCs
4	Species that breaks down ozone				Cl• radicals
5	Overall equation for ozone breakdown				$2O_3 \rightarrow 3O_2$
6	Why one Cl• destroys many O_3				Acts as a catalyst (regenerated)
7	Replacement for CFCs				HFCs
8	Why HFCs do not deplete ozone				C–F bond too strong to break
9	Nitrile functional group				–C≡N
10	Why excess NH_3 is used				Reduces further substitution

Fold the Answer column behind the page. Answer each question from memory, then unfold and check. Repeat on a later day in the next Attempt column.